

```

%% =====
% Shannon Capacity Calculation for Noisy Channel
% Author : Suraj Jha
% Description:
% This program generates a message signal, adds Gaussian noise,
% calculates SNR, estimates bandwidth, and computes
% Shannon Channel Capacity.
%% =====

clc;
clear;
close all;

%% =====
% SIGNAL GENERATION
%% =====

fs = 8000;           % Sampling Frequency (Hz)
t = 0:1/fs:1-1/fs;  % Time Vector

A = 2;               % Amplitude
f = 3;               % Signal Frequency

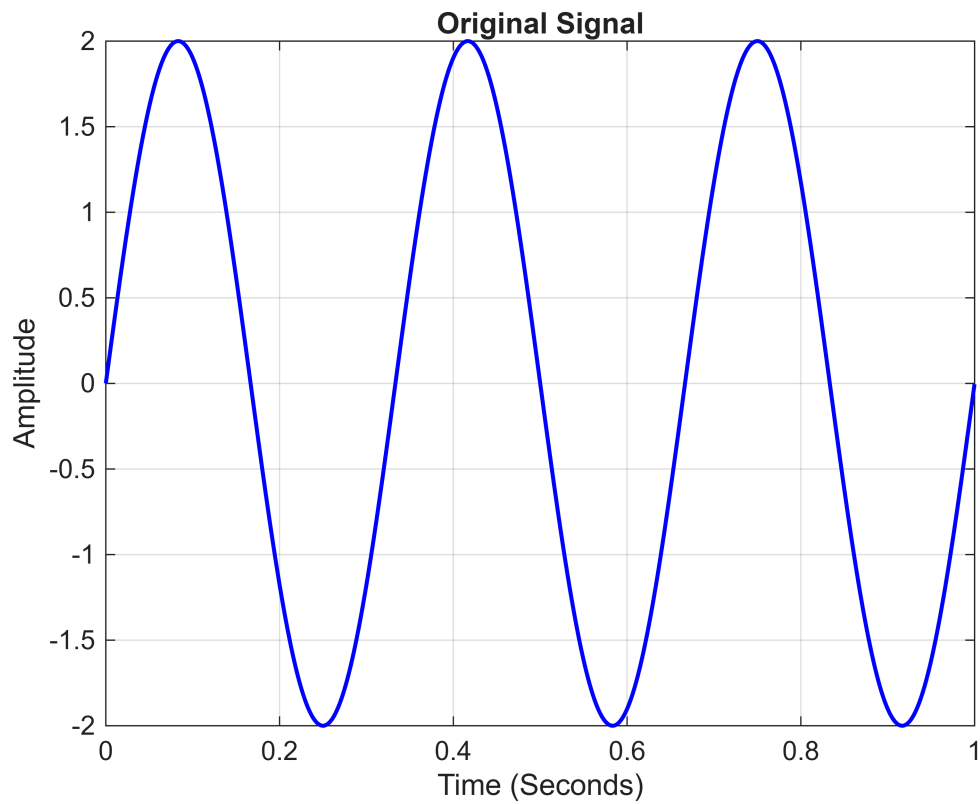
signal = A*sin(2*pi*f*t);

%% =====
% ORIGINAL SIGNAL
%% =====

figure;
plot(t,signal,'b','LineWidth',1.5);
grid on;
xlabel('Time (Seconds)');
ylabel('Amplitude');
title('Original Signal');

exportgraphics(gcf,'Figure1_OriginalSignal.png','Resolution',300);

```

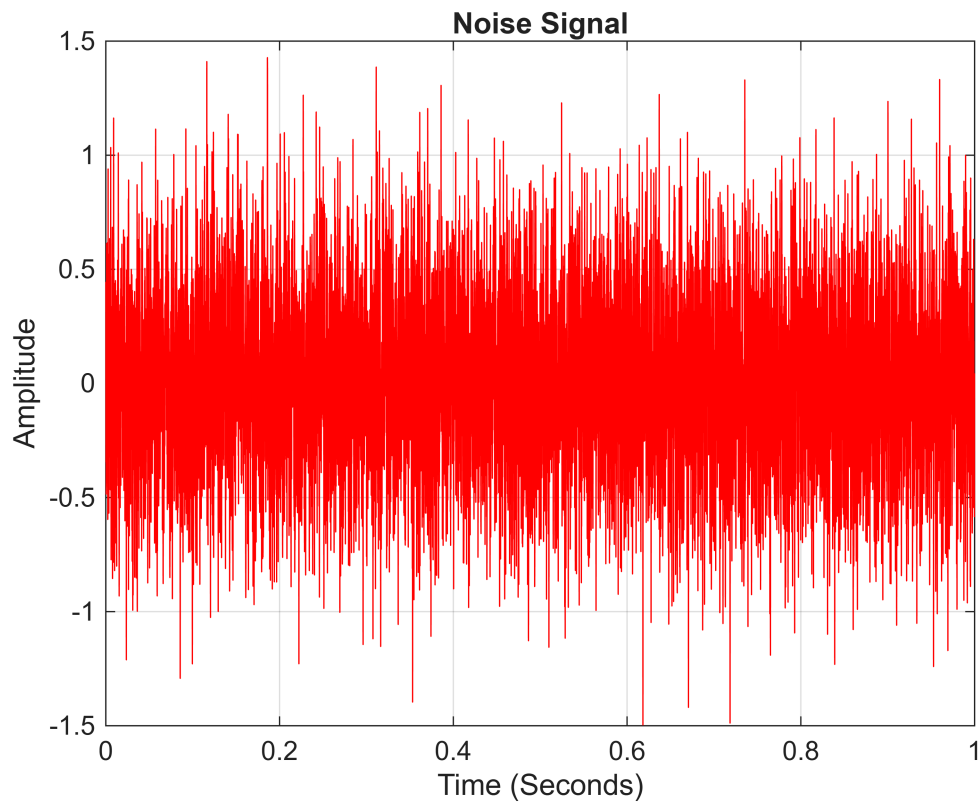


```
%% =====
% ADDITIVE WHITE GAUSSIAN NOISE
%% =====

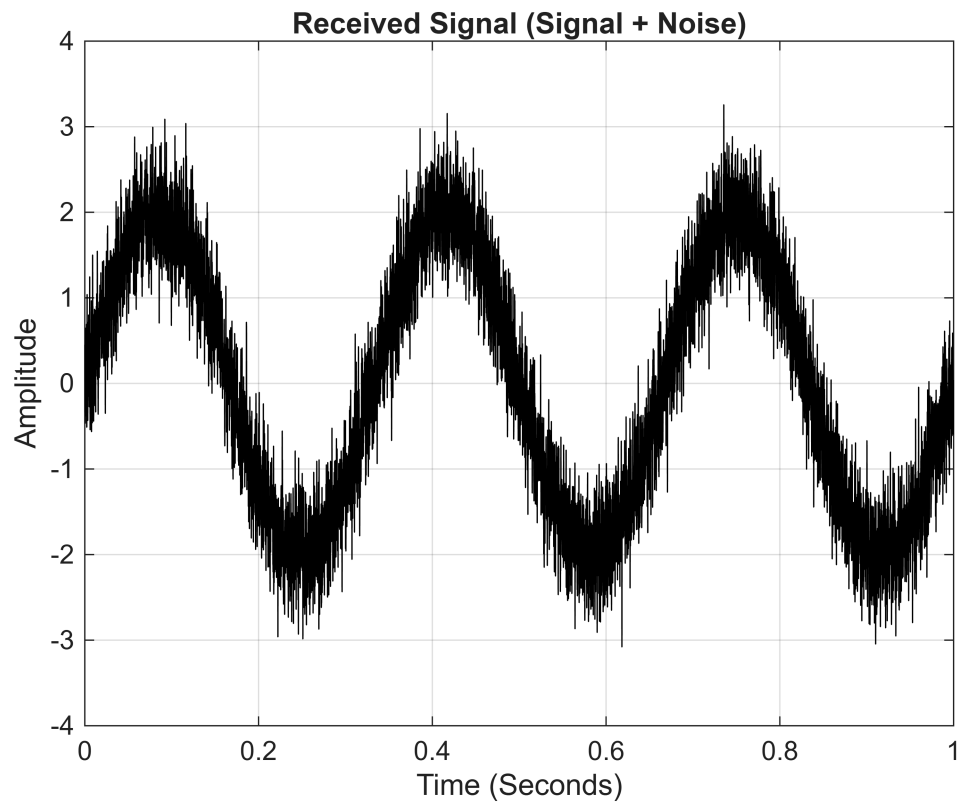
noise = 0.4*randn(size(signal));

figure;
plot(t,noise,'r');
grid on;
xlabel('Time (Seconds)');
ylabel('Amplitude');
title('Noise Signal');

exportgraphics(gcf,'Figure2_NoiseSignal.png','Resolution',300);
```



```
%% =====  
% SIGNAL + NOISE  
%% =====  
  
received = signal + noise;  
  
figure;  
plot(t,received,'k');  
grid on;  
xlabel('Time (Seconds)');  
ylabel('Amplitude');  
title('Received Signal (Signal + Noise)');  
  
exportgraphics(gcf,'Figure3_ReceivedSignal.png','Resolution',300);
```



```
%% =====
% SNR CALCULATION
%% =====
```

```
signalPower = mean(signal.^2);
noisePower  = mean(noise.^2);

SNR = 10*log10(signalPower/noisePower);

fprintf('Signal Power = %.4f W\n',signalPower);
```

```
Signal Power = 2.0000 W
```

```
fprintf('Noise Power  = %.4f W\n',noisePower);
```

```
Noise Power  = 0.1574 W
```

```
fprintf('SNR          = %.2f dB\n',SNR);
```

```
SNR          = 11.04 dB
```

```
%% =====
% BANDWIDTH CALCULATION
%% =====
```

```
bandwidth = obw(signal,fs);
```

```
fprintf('Bandwidth = %.2f Hz\n',bandwidth);
```

Bandwidth = 0.99 Hz

```
%% =====  
% SHANNON CAPACITY  
%% =====
```

```
SNR_linear = 10^(SNR/10);
```

```
Capacity = bandwidth * log2(1 + SNR_linear);
```

```
fprintf('Channel Capacity = %.2f bits/sec\n',Capacity);
```

Channel Capacity = 3.74 bits/sec

```
%% =====  
% FREQUENCY SPECTRUM  
%% =====
```

```
N = length(signal);
```

```
F = fs/2 * linspace(-1,1,N);
```

```
SignalSpectrum = abs(fftshift(fft(signal)))/(N/2);
```

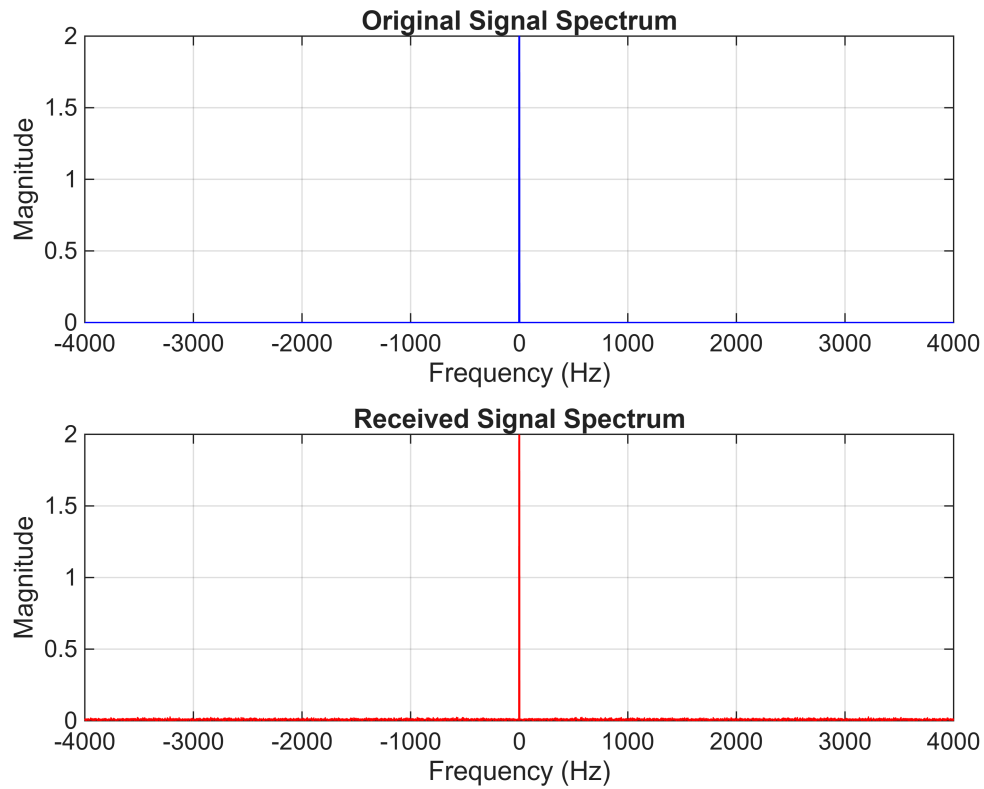
```
ReceivedSpectrum = abs(fftshift(fft(received)))/(N/2);
```

```
figure;
```

```
subplot(2,1,1)  
plot(F,SignalSpectrum,'b')  
grid on  
xlabel('Frequency (Hz)')  
ylabel('Magnitude')  
title('Original Signal Spectrum')
```

```
subplot(2,1,2)  
plot(F,ReceivedSpectrum,'r')  
grid on  
xlabel('Frequency (Hz)')  
ylabel('Magnitude')  
title('Received Signal Spectrum')
```

```
exportgraphics(gcf,'Figure4_Spectrum.png','Resolution',300);
```



```
%% =====
% DISPLAY RESULTS
%% =====
```

```
disp('-----');
```

```
disp(' Shannon Capacity Result');
```

```
Shannon Capacity Result
```

```
disp('-----');
```

```
fprintf('Bandwidth      : %.2f Hz\n',bandwidth);
```

```
Bandwidth      : 0.99 Hz
```

```
fprintf('SNR              : %.2f dB\n',SNR);
```

```
SNR              : 11.04 dB
```

```
fprintf('Channel Capacity: %.2f bits/sec\n',Capacity);
```

```
Channel Capacity: 3.74 bits/sec
```

```
disp('-----');
```

```
%% =====  
% END OF PROGRAM  
%% =====
```